

## Problem 1

In this problem, we use the OnlineNews dataset available on Canvas. The dataset is about predicting the popularity of websites from 58 features. Discard the first two variables from the data. From the remaining variables use all but the last variable as predictors and the log of the last variable.

Obtain all results using 5-fold cross-validation, which is computed as follows:

1. Generate a random permutation of the data. Use this randomized permutation to split the data into 5 disjoint subsets of almost equal size.
2. For each fold  $i \in \{1, \dots, 5\}$ , train the model on all data except subset  $i$  and test it on subset  $i$ , obtaining test error  $\epsilon_i$ .
3. Obtain the final test error as the average of the 5 test errors  $\epsilon_i$  obtained above.

Report results for the following models:

- Null model. Report the average train and test MSE of the null model that always predicts training  $\bar{y}$  (average training  $y$ )
- OLS regression computed analytically by solving the following normal equations:
- OLS regression with backward elimination, for example using SequentialFeatureSelector. For each number of features  $k \in \{5, 10, 15, \dots, 55\}$  report the average training and test  $R^2$  and MSE in a table. On the same graph, plot the average training and test  $R^2$  vs  $k$  as two separate curves
- Random forest regression with 100 trees and maximum depths  $1, 2, \dots, 10$ . Report the average training, OOB, and test  $R^2$  and MSE and their standard deviations in each case. Also, plot the average training, OOB, and test  $R^2$  vs the maximum depth as three separate curves.